

ASSIGNMENT 2

1. What is UML?

UML (Unified Modeling Language) is a standardized, general-purpose visual modeling language used in software engineering to specify, visualize, construct, and document the structure and behavior of a software system. It provides a common set of diagram types and notations so that developers, designers, and stakeholders can understand a system's design regardless of the programming language used to build it.

2. What is the use of UML?

UML is used to model a software system before and during development. It helps a team plan the system's structure (what data/entities exist and how they relate), its behavior (how users interact with it and how processes flow), and its architecture, before writing code. It improves communication between team members, makes the design easier to review and modify, and serves as documentation for future maintenance. For the AI Chatbot for College project, UML diagrams were used to plan how students and the admin interact with the system, and how the database entities relate to each other.

3. Explain types of UML diagram with example.

UML diagrams are broadly divided into two categories: Structural diagrams (show the static structure of a system) and Behavioral diagrams (show the dynamic behavior).

- **Class Diagram (Structural)** – Shows classes, their attributes, methods, and relationships. Example: in this project, a Student class with attributes roll_no, name, attendance, fees_due, course, semester, cgpa, and mentor.
- **Use Case Diagram (Behavioral)** – Shows actors and the actions (use cases) they can perform with the system. Example: a Student actor performing “Ask FAQ”, “Voice Input”, and “Login” use cases.
- **Sequence Diagram (Behavioral)** – Shows the order of messages exchanged between objects over time. Example: Student sends query → Flask backend queries SQLite → response back to Student.
- **Activity Diagram (Behavioral)** – Shows the flow of control/steps in a process, similar to a flowchart. Example: decision logic determining whether to fetch personal records or match general FAQs.
- **Component Diagram (Structural)** – Shows how software modules are organized. Example: Frontend UI component, Flask backend component, and SQLite database component.
- **Deployment Diagram (Structural)** – Shows the physical/hardware setup a system runs on. Example: Client browser → Render cloud Linux container → SQLite storage.

4. Draw UML Diagrams of your project.

(a) Use Case Diagram

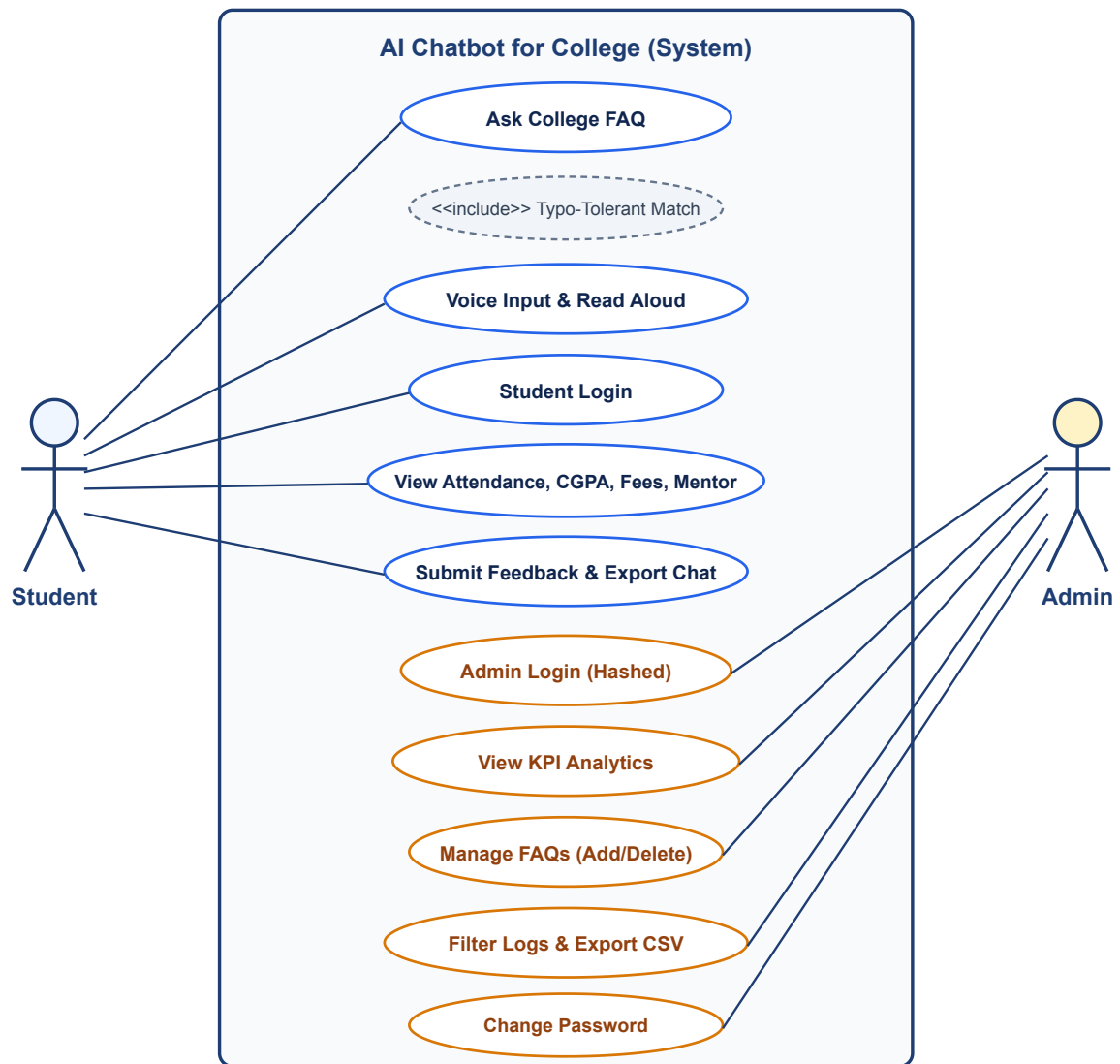


Fig 1: Use Case Diagram — Student and Admin actors and their interactions with the system.

(b) Class Diagram

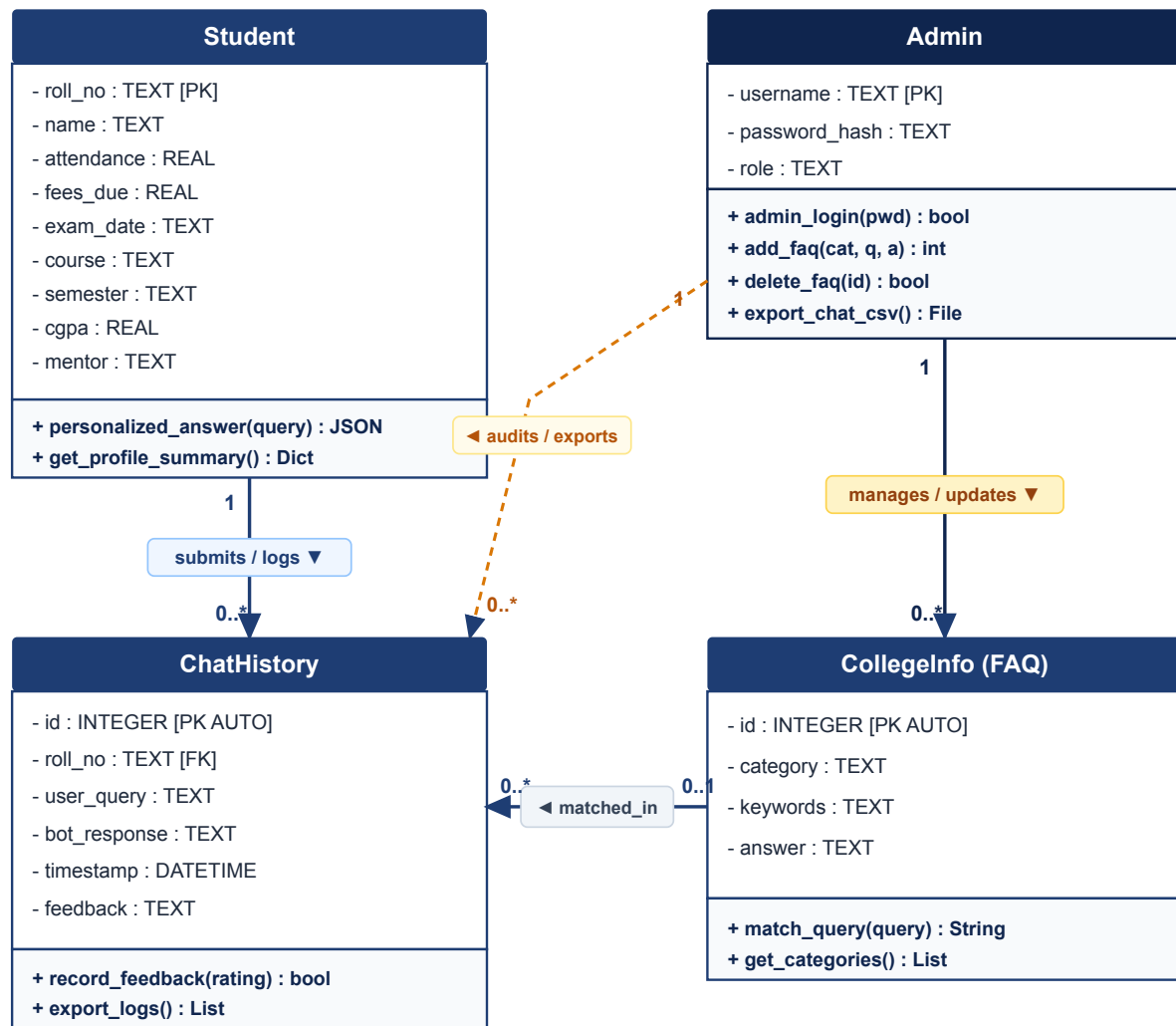


Fig 2: Class Diagram — fully connected system classes with associations, multiplicities (1 to 0..*), and role verbs.